Chapter 7
Industrial Calibration



7.1 Calibration Concept

According to VIM—2.39, calibration differs from adjustment and verification. See the calibration definition:

Operation that, under specified conditions, in a first step, establishes a relation between the quantity values with measurement uncertainties provided by measurement standards and corresponding indications with associated measurement uncertainties and, in a second step, uses this information to establish a relation for obtaining a measurement result from an indication.

NOTE 1 A calibration may be expressed by a statement, calibration function, calibration diagram, calibration curve, or calibration table. In some cases, it may consist of an additive or multiplicative correction of the indication with associated measurement uncertainty.

NOTE 2 Calibration should not be confused with adjustment of a measuring system, often mistakenly called "self-calibration," nor with verification of calibration.

NOTE 3 Often, the first step alone in the above definition is perceived as being calibration.

Note that, according to VIM, calibrating is confronting the metrological behavior of a measuring instrument with a reference standard, which can be a standard measurement instrument, a standard measurement system, a materialized measure, or a certified reference material.

Common sense considers calibrating how to fix equipment. For this reason, many professionals misunderstand that it is unnecessary to calibrate new equipment by thinking it is in perfect condition. The equipment may be in perfect condition, but we do not know its metrological characteristics, such as measurement error, measurement uncertainty, instrumental bias, and hysteresis.

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7.2 Calibration × Verification

VIM—2.44 defines verification as the "*provision of objective evidence that a given item fulfills specified requirements.*"

These specified requirements can be:

- The manufacturer's specifications;
- The measurement instrument hysteresis or linearity;
- The instrument measuring error compared to its specification or technical standard.

Example 7.1 Maximum Permissible Error for Resistance Thermometer
The maximum permissible error for class A and class B resistance thermometers, according to DIN-IEC 751/85 standard, is worth:

Class A: (0.15 + 0.002 T) °C.
Class B: (0.30 + 0.005 T) °C.

Where T is the value of the measurement temperature.

Example 7.2 Maximum Permissible Error for Materialized Measure
The maximum permissible error for a materialized measure, for example, a standard mass, depending on its class, according to the International Recommendation OIML R-111-1, is presented in Table 7.1.

Attention!
Do not confuse verification with calibration. In calibration, the measuring uncertainty of the object must be determined, but this is not necessary for verification. A calibration may cover a verification, but the opposite is not valid. In this sense, a calibration becomes a more complex procedure than a verification.

7.3 Measurement Standard

According to VIM—5.1, the measurement standard is the realization of the definition of a given quantity, with a *stated quantity value and associated measurement uncertainty, used as a reference.*

EXAMPLE 1 1 kg mass measurement standard with an associated standard measurement uncertainty of 3 μg.
EXAMPLE 2 100 Ω measurement standard resistor with an associated standard measurement uncertainty of 1 μΩ.
EXAMPLE 3 Caesium frequency standard with a relative standard measurement uncertainty of 2×10^{-15}.

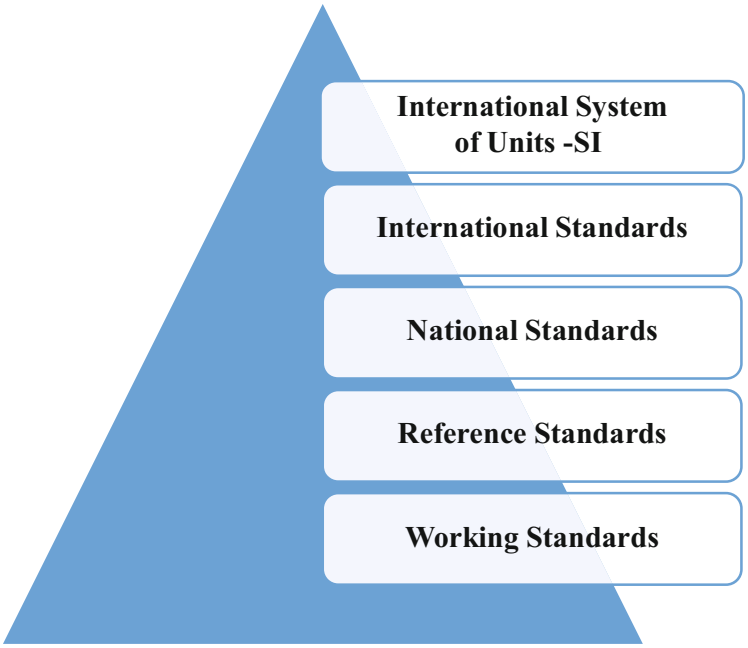


Fig. 7.1 Hierarchy of measurement standards

measurement standards often have a small measurement uncertainty, contributing little or almost nothing to the uncertainty after a calibration process. Their measurement uncertainty should be combined with the other measurement uncertainties involved in the calibration process.

A measurement standard serves as a reference for calibrating other hierarchically inferior standards in terms of accuracy and measurement uncertainty. Figure 7.1 presents the various types of measurement standards in the decreasing sense of their metrological hierarchy.

The International System of Units (SI) was presented and discussed in Chap. 1.

7.3.1 International Measurement Standard

In the VIM—5.10, the definition of an **international measurement standard** is a *measurement standard recognized by signatories to an international agreement and intended to serve worldwide.*

EXAMPLE 1 The international prototype of the kilogram. (Authors note: At the time of the 3rd edition of VIM, 2012, this was still the definition of the international mass standard—kilogram. From 2019, the new definition is: "It is defined by taking the fixed numerical value of the Planck constant h to be $6.626070\ 15 \times 10^{-34}$ when expressed in the unit $J\ s$, which is equal to $kg\ m^2\ s^{-1}$, where the meter and the second are defined in terms of c and $\Delta\nu_{Cs}$.")

EXAMPLE 2 Chorionic gonadotrophin, World Health Organization (WHO) 4th international standard 1999, 75/589, 650 International Units per ampoule.

EXAMPLE 3 VSMOW2 (Vienna Standard Mean Ocean Water) distributed by the International Atomic Energy Agency (IAEA) for differential stable isotope amount-of-substance ratio measurements.

Knowing a Little More ...
The standard kilogram

King Louis XVI of France summoned a group of sages to elaborate a new measurement system, establishing the foundations for the "decimal metric system," which evolved into modern SI. The original idea of the King's commission (which included remarkable, like Lavoisier) was to create a mass unit that, by definition, would be the mass of a liter of water at the ice point (i.e., essentially 1 kg). The definition should be incorporated into a standard mass prototype. Given that the masses being measured at that time were much lower than the kilo, they decided that the unit of mass would be the "gram." However, since a gram standard is difficult to use as well as to handle (too small), the new definition should be incorporated into a prototype of a kilogram. The republican government's decision probably had political motivation, after all, these same people condemned Lavoisier to the guillotine. Anyway, it remains to regret that a base unit has a "prefix" in the name. The kilogram prototype has been conserved in the BIPM since 1889, when it was sanctioned by the First General Conference on Weights and Measures. It is cylindrically, with a diameter and height of about 39 mm, and made of a 90% platinum alloy and 10% iridium. In November 2018, at the General Conference on Weights and Measures, the highest revision of the International Unit System (SI) was made since 1960. In this conference, four basic units of measurement were redefined: kilogram, ampere, kelvin, and mol. The purpose of the change was to relate these units to fundamental, not arbitrary constants, as has been done so far.

7.3.2 National Measurement Standard

In the VIM—5.3, the definition of a **national measurement standard** is a *measurement standard recognized by national authority to serve in a state or economy as the basis for assigning quantity values to other measurement standards for the kind of quantity concerned.*

Therefore, these standards are devices maintained by organizations and national laboratories worldwide. They represent the fundamental and derived quantities and are calibrated independently through absolute measurements.

INMETRO is responsible for maintaining national standards in Brazil both for those existing in their laboratories and those in the designated laboratories.

7.3.3 Reference Measurement Standard

In VIM—5.6, a **reference measurement standard** is a *measurement standard designated for the calibration of other measurement standards for quantities of a given kind in a given organization or at a given location.*

These standards should not be employed for daily measurement work and, preferably, have to be kept under specific conditions of temperature and humidity. They are used for the calibration or verification of working standards.

7.3.4 Working Measurement Standard

The definition of a **working measurement standard** in VIM—5.7 is a standard that is *routinely used to calibrate or verify measuring instruments or systems.*

NOTE 1 A working measurement standard is usually calibrated with respect to a reference measurement standard.

NOTE 2 In relation to verification, the terms "check standard" or "control standard" are also sometimes used.

7.4 Certified Reference Material (CRM)

Certified reference material is a *reference material, accompanied by documentation issued by an authoritative body and providing one or more specified property values with associated uncertainties and traceabilities, using valid procedures.*

EXAMPLE Human serum with assigned quantity value for the concentration of cholesterol and associated measurement uncertainty stated in an accompanying certificate, used as a calibrator or measurement trueness control material.

NOTE 1 'Documentation' is given in the form of a 'certificate' (see ISO Guide 31: 2000).

NOTE 2 Procedures for the production and certification of certified reference materials are given, e.g.*, in ISO Guide 34* (actually ISO 17034) *and ISO Guide 35.*

NOTE 3 In this definition, "uncertainty" covers both 'measurement uncertainty' and 'uncertainty associated with the value of a nominal property,' such as for identity and sequence. "Traceability" covers both 'metrological traceability of a quantity value' and 'traceability of a nominal property value.'

NOTE 4 Specified quantity values of certified reference materials require metrological traceability with associated measurement uncertainty.

NOTE 5 ISO/REMCO has an analogous definition, but uses the modifiers "metrological" and "metrologically" to refer to both quantity and nominal property. [VIM—5.14].

A reference material can be a pure substance or a gas, liquid, or solid mixture. Examples of reference materials are water used in viscometer calibration, sapphire used in calorimetry heat capacity calibration, and solutions used in chemical analysis.

A certificate of analysis with one or more property or physical characteristic values always accompanies a certified reference material. These materials are certified by procedures that establish the traceability for the exact obtaining of the unit in which the property values are expressed. Each certified value is accompanied by uncertainty to an established confidence level.

The following text, related to the preparation of a CRM, was obtained from the website <https://en.wikipedia.org/wiki/Certified_reference_materials>.

Sample preparation.
Detailed sample preparation depends on the type of material. Pure standards are most likely prepared by chemical synthesis and purification and characterized by determining remaining impurities.

Natural matrix CRMs contain an analyte or analytes in a natural sample (for example, lead in fish tissue). They are typically produced by homogenizing a naturally occurring material and measuring each analyte. Due to the difficulty in production and value assignment, they are usually made by national or transnational metrology institutes like NIST (USA), BAM (Germany), KRISS (Korea), and EC JRC (European Commission Joint Research Centre).

Natural materials are rarely homogeneous on the scale of grams so production of a solid natural matrix reference material typically involves processing to a fine powder or paste. Homogenization can have adverse effects, for example on proteins, so producers must take care not to over-process materials.

The stability of a certified reference material is also essential, so a range of strategies may be used to prepare a reference material that is more stable than the natural material it is prepared from. For example, stabilizing agents such as antioxidants or antimicrobial agents may be added to prevent degradation, liquids containing certified concentrations of trace metals may have their pH adjusted to keep metals in solution, and clinical reference materials may be freeze-dried for long-term storage if they can be reconstituted successfully.

7.5 Selection of the Measurement Sstandard

For the value of a measurement standard or standard measurement system to be accepted as a reference value, its measurement accuracy and uncertainty must be lower than those of the measurement system to be calibrated. Therefore, it is possible

We observe that the influence of *SMS* uncertainty is approximately 3.2% of the final uncertainty, U_F.

(e) Considering $U_{\mathrm{SMS}} = 1/5\ U_{\mathrm{CMS}}$, we have:

$$\boldsymbol{U_F = \sqrt{U_{CMS}^2 + U_{SMS}^2}} = \sqrt{(1 + 1/25)\ U_{CMS}^2} = 1.02\ U_{CMS}$$

We observe that the influence of *SMS* uncertainty is approximately 2% of the final uncertainty, U_F.

(f) Considering $U_{\mathrm{SMS}} = 1/10\ U_{\mathrm{CMS}}$, we have:

$$\boldsymbol{U_F = \sqrt{U_{CMS}^2 + U_{SMS}^2}} = \sqrt{(1 + 1/100)\ U_{CMS}^2} = 1.005\ U_{CMS}$$

We observe that the influence of *SMS* uncertainty is approximately 0.5 % of the final uncertainty, U_F.

Suppose we adopt a standard with measurement uncertainty equal to or less than one-tenth of the expected uncertainty for the *CMS*. In that case, *SMS* will not contribute to the uncertainty, with *SMS* passing unnoticed by the *CMS*.

Important
In practice, if we adopt *SMS* with measurement uncertainty equal to or less than ¼ of the *CMS*, we will have an excellent condition.

7.6 Solved Exercises of Measurement Instrument Calibration

We can use a technical standard or even an orientation guide whenever we want to calibrate a measurement instrument. For example, if we wish to calibrate thermocouples, we can use the guide *Calibration of Thermocouples—EURAMET cg-8*.

If we wish to calibrate manometers, we can use the *Guidelines on the Calibration of Electromechanical and Mechanical Manometers—EURAMET Calibration Guide No. 17*.

Thus, we are led to believe that the entire calibration process, including the choice of standards, the assembly of the experiment, and the calculation of the measurement uncertainty, will be found in all technical standards. Deceit! The part of the measurement uncertainty calculation is not usually provided in the technical calibration standards, so we are at a "dead end."

Table 7.19 Metrological characteristics

Metrological characteristics	(mV)
Object	200.00
Error	−0.26
Type A uncertainty—Repeatability	0.0138
Standard uncertainty from certificate	0.001
Parasite uncertainty	0.0011547
Object resolution uncertainty	0.002887
Standard resolution	0.0002887
Combined uncertainty	0.0142
Effective degree of freedom	3.4
Coverage factor, k	3.31
Expanded uncertainty, 95.45 %	0.05

Table 7.20 Relative error

Point (mV)	Error (mV)	Error (%)
40.00	−0.16	0.08
80.00	−0.16	0.08
120.00	−0.19	0.10
160.00	−0.18	0.09
200.00	−0.26	0.13

Table 7.21 Result of calibration

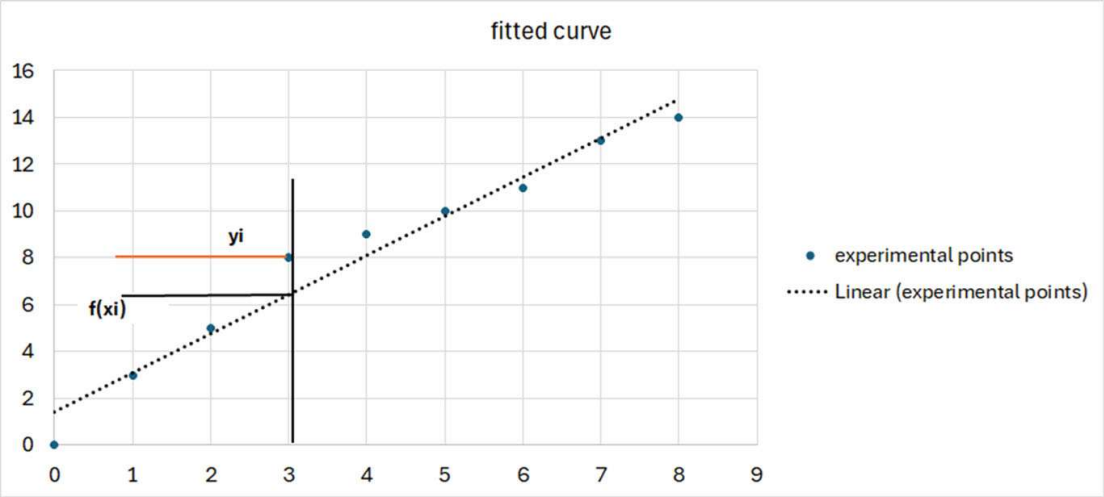
Object mV	Error mV	Uncertainty mV	k	υ
40.00	−0.16	0.04	3.31	3.5
80.00	−0.16	0.03	3.31	4
120.00	−0.19	0.03	3.31	3.7
160.00	−0.18	0.05	3.31	3.4
200.00	−0.26	0.05	3.31	3.4

7.7 Measurement Uncertainty in Fitting a Function

Most of the time, a phenomenon or a physical, chemical, or mechanical process is represented by an experimental dataset. In these cases, it can be extremely interesting to "represent" this dataset by a defined mathematical function. This approach procedure is known as fitting or a function regression, and one of the techniques used is the **Least-Squares Method** (LSM).

The literature widely discusses the fitting of experimental points by the LSM, and we usually adopt software that makes these adjustments, such as Microsoft Excel©. For this reason, we do not intend to address the demonstrations of the equations that allow the determination of both the fitting function and its uncertainties.

We are interested in presenting the technique for calculating the measure uncertainty of a function by the LSM. We apply this method whenever we want to describe experimental data behavior—for example, the results in a calibration certificate—through a mathematical equation.



Graph 7.1 The adjustment of experimental points

This method consists of adjusting the dataset to a function that minimizes the experimental variance of the set, that is, we must reduce the difference:

$$f(x_i) - y_i$$

Where $f(x_i)$ is the value of the fitted function for point x_i, and y_i is the experimental value obtained for point XI, as shown in Graph 7.1.

As the method minimizes the difference but does not eliminate it, we will always have to fit a function, whether it is the first (linear), of the second degree (parable), or any other order, to obtain a measurement uncertainty related to both coefficients of this function as those of its value on the y axis.

7.7.1 *Measurement Uncertainty of* Y

When we fit an experimental curve, for example, when we make a calibration curve relating to the *y-axis*, the value of the standard, and the *x-axis*, the value of the calibrated object, we generate a function $f(x_i)$ with measurement uncertainty associated with fitting, since no adjustment is perfect.

According to GUM 2008—Guide to the expression of uncertainty in measurement (Annex H.3.2 Least-Squares Fitting), the variance s^2 is a measure of the overall uncertainty, and the equation that determines the fitting uncertainty of the values found on the axis y is as follows:

$$u_{\text{fitting}} = \sqrt{s^2} = \sqrt{\frac{1}{n-p}\sum_1^n [f(x_i) - y_i]^2} \tag{7.2}$$

Where $f(x_i)$ is the value of the fitting function for point x_i; y_i is the experimental value obtained for point x_i; p is the number of parameters to be fitted; n is the number of experimental points; and $(n - p)$ is the fitting degree of freedom.

7.7.2 Fitting Uncertainty

Considering the experimental points (x, y) obtained through calibration and using a standard instrument, the total uncertainty of variable y will be the combination of the uncertainty of the calibrated object (u_{object}) and the fitting uncertainty (u_{fitting}) through the equation:

$$u_y = \sqrt{u_{\text{fitting}}^2 + u_{\text{object}}^2} \tag{7.3}$$

Fitting uncertainty and the calibrated object uncertainty should always be combined in a standardized form.

Solved Exercise 7.4: Calibration Graph—Fitting the Experimental Data.

It should be a digital scale with a measurement range of 0 to 50.00 kg and a resolution of 0.01 kg. Table 7.22 shows the calibration result.

We know the uncertainties at each point of the scale. Suppose we now want to obtain an equation that describes the behavior of the scale at any point within the calibration interval 0 to 50 kg. Make the calibration curve of this scale and determine the final uncertainty considering the fitting made by a first-degree function.

Solution Adjusting the scale calibration points by a line of type $y(x) = ax + b$, we will have Graph 7.2:

Table 7.22 Calibration result

Standard (kg)	Object (kg)	Bias (kg)	Uncertainty (kg)	k
0.000	0.02	0.02	0.01	2.00
5.000	4.97	− 0.03	0.01	2.04
10.000	10.02	0.02	0.01	2.04
15.000	14.96	− 0.04	0.01	2.08
20.000	20.02	0.02	0.01	2.09
25.000	24.98	− 0.02	0.01	2.09
30.000	30.05	0.05	0.02	2.09
35.000	35.01	0.01	0.02	2.05
40.000	39.99	− 0.01	0.02	2.06
45.000	45.02	0.02	0.03	2.09
50.000	49.99	− 0.01	0.03	2.09

Table 7.29 Point 70.0 °C

Metrological characteristics	Results
Mean electric current (mA)	15.22
Mean temperature standard (°C)	70.30
Object temperature (°C)	70.3
Sensitivity coefficient (°C/mA)	6.26721950
Type A uncertainty—Object repeatability (°C)	0.00
Uncertainty from the standard certificate (°C)	0.025
Multimeter uncertainty (°C)	0.41288442
Bath stability uncertainty (°C)	0.0288675
Fitting uncertainty (°C)	0.253848768
Combined uncertainty (°C)	0.486180084
Effective degree of freedom, υ_{eff}	53
Coverage factor, k	2.05
Expanded uncertainty, 95.45 % (°C)	1.0

Table 7.30 Point 90.0 °C

Metrological characteristics	Results
Mean electric current (mA)	18.30
Mean temperature standard (°C)	90.00
Object temperature (°C)	89.6
Sensitivity coefficient (°C/mA)	6.26721950
Type A uncertainty—Object repeatability (°C)	0.00
Uncertainty from the standard certificate (°C)	0.025
Multimeter uncertainty (°C)	0.490096564
Bath stability uncertainty (°C)	0.0288675
Fitting uncertainty (°C)	0.253848768
Combined uncertainty (°C)	0.553255973
Effective degree of freedom, υ_{eff}	90
Coverage factor, k	2.03
Expanded uncertainty, 95.45 % (°C)	1.1

IV. Point 70.0 °C (Table 7.29)

V. Point 90.0 °C (Table 7.30)

VI. Point 100.0 °C (Tables 7.31 and 7.32)

7.8 Proposed Exercises

7.8.1 The calibration of a temperature sensor (PT-100 to 3 wires) against a temperature standard presented the values contained in Table 7.33